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Model-based predictive digital twin for bridge structural health monitoring: the integration of building information modeling, finite element analysis, and machine learning in a Netherlands case study

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A R T I C L E   I N F O

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A B S T R A C T

Advancements in bridge health monitoring systems are a trending issue for many researchers and industry professionals these days, where the bridge community is looking for Artificial Intelligence (AI) based automated solutions. To address this challenge, this study integrates Building Information Modeling (BIM), Finite Element Method (FEM), and Machine Learning (ML) algorithms to create a comprehensive Digital Twin (DT) framework capable of real-time monitoring and predictive analysis of the bridge's structural integrity. A robust design of the workflows and technical framework is proposed to enable seamless interaction between the physical bridge assets and the proposed model-based Predictive Digital Twin (PDT) system using virtual designs and Neural Network (NN) prediction algorithms. The major research contributions of this article include model-based Bridge Structural Health Monitoring (BSHM) using accurate predictions from ML and sensor data and the novel PDT-based approach. It proposes information flows through an integrated BIM, ML, and DT platform to ensure improved BSHM. The proposed ML-based PDT framework is validated using a steel girder bridge located in the Netherlands, which demonstrates automated health monitoring of the bridge. The PDT system is validated using actual BSHM data (strain data), with predictions from the ML model compared against known load cases. The outcomes of this research showcase the PDT framework's ability to predict load placement and magnitude with precision, contributing to the ongoing advancement of suitable infrastructure management practices.

1. Introduction

Bridges are among some of the most critical infrastructures that facilitate transportation and logistics. The importance of these infrastructures and the severe consequences of their failures have motivated an increased interest in the timely assessment of these structures. The partial collapse of the Seongsu Bridge in South Korea, which resulted in the deaths and injuries of 32 and 17 people, respectively (Reuters, 1994), was one of the earliest incidents that increased interest in bridge health monitoring (BHM) systems. Later incidents, such as the failures of four bridges in Italy in 2014 [2], 2016 [3], 2017 (Bazzucchi and Ferro, 2019), and 2018 (Calvi et al., 2019), further heightened the necessity of these BHM systems. Some of the earliest BHM systems were introduced as early as the 20th century (Eth-bibliothek et al.). For example, the Versoix Bridge in Switzerland was one of the first large-scale BHMs that utilized 100 fiber optic sensors in 1996 (Inaudi et al., 2016). Unfortunately, events such as the terrorist attack on the World Trade Towers in New York City in 2001 slowed the progress of Structural Health Monitoring (SHM) as many asset owners in the US decided against sharing data about their structures to conceal their possible weaknesses from terrorist attacks. However, the research projects in Europe and the Far East continued to advance the field of BHM

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systems. Particularly, Europe and Italy have initiated some large-scale projects in this field in recent years (Aktan et al., 2024; Bazzucchi et al., 2018; Di Prisco et al., 2023).

In traditional BHMs, numerous sensors are installed on different parts of the bridge to track structural responses, including strain, deflection, and vibration. Parameters, such as ambient temperature and weather conditions, vehicular load, and external impacts may be additionally measured through the sensor network. These systems collect measurements, analyze them, and detect anomalies in the behavior of the bridge, enabling early warnings and proactive maintenance (Wan et al., 2023). However, these early systems suffered from shortcomings caused by their analysis techniques and sensor networks. For example, a study (Robertson, 2005) showed that some widely used prediction models were inaccurate. Advancements in sensor technology and analysis methods have improved modern BHMs. Wireless sensors, remote sensing, and imaging technologies such as LiDAR, Unmanned Aerial Vehicles (UAVs), and robots, have expanded the research horizon on monitoring capabilities. In parallel, improvements in data communication and analytical tools have facilitated their utilization (Mahmud et al., 2018). In particular, computer vision-based and AI-based BHMs have been on the rise in recent years (Zhang et al., 2022a) (Zinno et al., 2022).

Different AI architectures and methods have been used in SHM systems to gain new insights or to replace previous methods. Anomaly detection using computer vision-based deep learning methods is one of these applications (Zhang et al., 2022a). Other research utilized computer vision-based deep learning to monitor the traffic load on bridges (Ge et al., 2020). In BHM systems, different circumstances, such as sensor failure, can cause a loss of data, which in turn can have serious consequences in the later stages of the process. Thus, data reconstruction is necessary to improve the BHM systems. A deep learning framework was proposed to recover the missing wind data of bridges (wei Wang et al., 2022). In another study, a Convolutional Neural Network (CNN)-based network was used to recover missing vibration data (Fan et al., 2019). Similarly, Zhang et al. (2022b) utilized Generative Adversarial Networks (GAN) and their variations to interpolate and predict the missing data in BHM systems. Another field in which AI has been widely applied to BHM systems is the identification of damage. Artificial Neural Networks (ANNs) were applied to identify and predict possible damage by comparing predicted and actual values (Malekjafarian et al., 2019). ANNs based on vibration-based methods were the subject of multiple studies (Sonbul and Rashid, 2023) (Nick et al., 2021) and were applied in BHMs to identify the possible damage to bridges. AI techniques can also be used to conduct and emulate Non-Destructive Testing (NDT). For example, thermal images can be used to identify corrosion and delamination in the steel surface of bridges by deep-learning methods (Ali and Cha, 2019).

Bridge management systems are mainly based on the manual inspection of the bridges by inspectors (Wan et al., 2019), and the data collected by them is in the form of written reports or fragmented digital data (Ciccone et al., 2022). This inefficiency increases the cost and time of the process and may cause risks to the safety and reliability of the bridge. This limitation may arise from communication errors, loss of information, difficulty in transferring information, or misunderstanding caused by the lack of a central information management system (Fawad et al., 2024a). These problems and other challenges have highlighted the necessity of digitalizing bridge inspection and management processes (Jahnke et al., 2023). Recently, Building Information Modeling (BIM) has revolutionized the construction sector as a means to digitalize buildings. However, BIMs are mostly applied to the design and construction phases of the buildings, while the operation and maintenance (O&M) phase has been mostly neglected (Fawad et al., 2023). A specialized variant of BIM is Bridge Information Modeling (BrIM), which is dedicated only to bridges. While BrIMs are also mostly applied to the design and construction phases, studies have shown that they can be applied to all phases of the bridge life cycle (Salamak, 2020). Some of the potential improvements from this transformation are leveraging data-driven methods to support maintenance decisions, thus increasing the efficiency of resource allocation and reducing their associated risks (Fawad et al., 2024b). BrIMs can be applied to the O&M phase of bridges by improving the inspection of bridges through BHM systems. In recent years, many studies have been conducted to integrate new technologies into the BrIMs. For example, a Mixed Reality (MR) application was leveraged to improve the inspection process by allowing the inspectors to conduct remote inspections and manage the maintenance process from a mixed-reality-enabled BrIM (Fawad et al., 2025).

2. Literature review

2.1. Advancements in bridge SHM method and workflows

Periodic inspections traditionally identify damage to civil engineering structures through Non-Destructive Evaluation (NDE) or visual inspection (Malekloo et al., 2022). However, these methods are impractical for large and complex civil engineering structures, such as bridges, and the interval between inspections can also present additional challenges (He et al., 2022). These issues have led to an increasing demand for Bridge Structural Health Monitoring (BSHM) methods (Gatti, 2019), enabling the transition from static damage assessments to real-time or online assessments (He et al., 2022) (Figueiredo E, 2022). Furthermore, infrastructure is aging and is increasingly unable to accommodate current traffic volumes. This situation necessitates more critical assessments and has contributed to the growing adoption of SHM techniques (Azhar et al., 2024). SHM can be defined as a set of strategies created to detect anomalies and damage at an early stage through sensors and monitoring systems that collect and analyze real-time data (Prendergast et al., 2018). The increasing need to maintain and ensure the structural integrity of infrastructures, coupled with technological advancements of the Industry 4.0 era, has driven the adoption of advanced technologies such as Building Information Modeling (BIM) (Angelosanti et al., 2023) (Gragnaniello et al., 2024), Virtual (VR) and Augmented Reality (AR) (Jahnke et al., 2023) (Fawad et al., 2022a), Finite Element Analysis (FEA) (Fawad et al., 2022b) (Fawad et al., 2019), and Machine Learning (ML) (Hielscher et al., 2023) (Sarker, 2021).

2.2. Development of digital twin systems for bridge monitoring

In recent years, the concept of Digital Twins (DT) has garnered significant attention for its potential to revolutionize SHM systems by enabling real-time monitoring, predictive analysis, and decision-making capabilities (Tuhaise et al., 2023) (Al-Hijazeen et al., 2023). DT, by integrating multiple data streams and modeling techniques, enables continuous assessment of structural conditions, thereby facilitating proactive maintenance and reducing the risk of unexpected failures (Smarsly et al., 2024) (Jiménez Rios et al., 2023). A generic DT in civil engineering can be defined as a digital representation of a physical entity that dynamically mirrors its physical counterpart and remains synchronized with it throughout at least part of its lifecycle (Tao et al., 2019) (Tuhaise et al., 2023).

Interoperability plays a crucial role in establishing a DT model for civil structures (Michael et al., 2024). It can be seen as the ability to use and exchange multi-source data and information between diverse components of a system (Naderi and Shojaei, 2023). The literature mentions several paradigms for the multi-elementality and modularity of DT frameworks. Liu et al. (2023a) distinguished three core components of DT, namely a physical entity, its virtual replicas, and twin data. Jia et al. (2022) suggested the division of DT components into three groups of different purposes: basic, visualization, and service components. By their concept, the basic components are responsible for data acquisition and processing. The visualization components, in turn, are used as the interface of the framework, enabling human-computer interaction. The last group, the service components, should be based on analytical tools for data analysis, including simulation, optimization, and prediction. A similar approach was incorporated by Li et al. (2024).

2.3. Integration of SHM, DT, and ML

The integration of SHM with DT enables an accurate representation of a structure's performance, as sensors continuously collect data that can be used to monitor the infrastructure's health (Chacón et al., 2023). Interaction between different components of DT can then be achieved using both data-driven and model-based methods. The data-driven SHM techniques, usually based on ML, are generally more effective at data processing and uncertainty handling, and they are typically faster during the inference stage than model-based approaches (Darko et al., 2020) (Wan et al., 2023). However, the choice between model-based and data-driven SHM is not always simple and depends on multiple factors. Incorporating ML models allows predictions regarding the structural behavior of bridges under various load conditions (Zhang et al., 2024a), enhancing the ability to detect early signs of damage or degradation (Teng et al., 2023) (Heng et al., 2024). Model-based methods, such as FEA, can further complement the DT, providing detailed simulations to support the system (Liu et al., 2023b) (Zhang et al., 2024b).

To simplify the user experience, data from SHM systems can be displayed on the DT model (Liu et al., 2023c). This requires combining and integrating models at several levels of preparation (Krishnamurthi et al., 2020). Fawad et al. (2024b) (Fawad et al., 2025) used the gaming technology based on the Unity engine to place virtual sensors on the BIM model. This approach allowed using AR and displaying sensor data in real-time and improved bridge maintenance management. Moreover, existing bridges do not always have a BIM model created. In this case, the use of point cloud-based techniques, namely scan-to-BIM, is investigated (Pepe et al., 2021) (Rausch and Haas, 2021) (Mafipour et al., 2023) (Shu et al., 2024). Model-based visualization provides the opportunity to display multiple datasets in one environment, which is easy and convenient to adapt to bridge management duties (Yang et al., 2024).

2.4. Research gap

Although many studies have examined the application of ML in SHM, and several different monitoring frameworks utilized combined FEA and BIM, the seamless integration of these components remains insufficiently explored. Thus, the novelty of this research lies in its approach to integrating BIM, FEA, and ML with SHM data through a custom-developed Revit add-in (API). This add-in enables seamless interaction between the BIM environment and SHM data, allowing for potentially automated SHM of the bridge. By leveraging FEA and sensor data, the novel ML algorithms develop the DT framework of the integrated system that not only provides insights into current structural performance but also emphasizes the data-driven predictive capabilities that can improve maintenance strategies and optimize resource allocation.

3. Research framework

This study focuses on developing a PDT dashboard for SHM systems of bridges and is tested on a steel girder bridge located at the University of Twente, Netherlands. The bridge, spanning 27 m and serving as a pedestrian and cyclist crossing, is equipped with an SHM system comprising various sensors, including strain gauges, accelerometers, and temperature sensors. The integration of BIM, FEA, and ML into a single PDT framework creates a robust platform for real-time monitoring and predictive maintenance, contributing to the ongoing advancement of sustainable infrastructure management practices (Saif et al., 2024). This research aims to validate the effectiveness of this PDT system by demonstrating the dashboard on a real bridge. The overall workflow is illustrated in Fig. 1.

The following sections outline the development and validation process of this PDT system and discuss its potential implications for infrastructure monitoring and management. Section 4 describes the software tools used in the study and explains the integration of BIM, FEA, and ML within the PDT framework. Autodesk Revit serves as the core BIM environment. Section 5 describes the bridge structure in detail, including its physical characteristics and sensor setup for monitoring. Section 6 presents the development of a custom Revit add-in and discusses the integration of the FEA model with the ML component. This section concludes by exploring the validation of the PDT through real-world data collected from the bridge. The findings and limitations of this study are discussed in Section 7, and the paper is concluded in Section 8.

4. Selected development toolkit

This research integrates Building Information Modeling (BIM), Finite Element Analysis (FEA), and 3D rendering software for visualization and validation of the performance of the structural system. The framework incorporates a BIM model of the bridge. The commercial software Autodesk Revit is used as the BIM environment (Autodesk, 2024). The DT framework is created in the form of an interfaced add-in, using the C# object-oriented programming language, which forms the basis of the Revit API. The tool runs as a Revit add-in accessible from the software ribbon.

In order to develop the Revit add-in, Visual Studio is adopted. Visual Studio is an IDE (Integrated Development Environment) and multi-language programming platform supporting object-oriented languages, including C# and Python (Microsoft Inc, 2024). The environment is used to complete the source code for the back-end and front-end of the Revit add-in.

In order to perform ML, Python-based libraries for various regression and classification problems are used (Python Software Foundation, 2024). The research assumes that one of the libraries is used to create an ML model for the prediction of the placement and weight of the load on the bridge using strain values measured by selected sensors.

Thus, by exploring the in-depth capabilities and adaptability of DTs for bridge health monitoring and their integration with BIM, this research contributes to the ongoing evolution of efficient and sustainable infrastructure management practices. The software used and the schematic of PDT development and its integration with ML predicted values are shown in Fig. 1.

5. Description of the bridge case study: University of Twente (UT) campus bridge

The 27 m-long steel girder bridge is located at the University of Twente (UT) campus in the Netherlands (Panigati et al., 2025). The UT Campus bridge connects Hogenkamp (East) and the sports stadium (West) across a pond. The bridge was possibly commissioned by the public in the 1980s as a pedestrian footbridge for pond crossing. Nowadays, it shows some signs of corrosion. It is positioned in the north-west direction and due to nearby trees, only the south-west side of the bridge is directly exposed to sunlight, particularly during the spring and autumn months. Thus, the west side of the case study is mostly hidden from the sun. This leads to different thermal expansion and contraction patterns in the structure (Voordijk and Kromanis, 2022) (Marchenko et al., 2024). The bridge structure is shown in Fig. 2.

The bridge is monitored using an SHM system consisting of distributed temperature sensors, accelerometers, inclinometers, strain gauges, and a local weather station (Cunha et al., 2021). These elements make it an interesting subject for research, promising valuable insights on the design of the SHM system, installation of sensors, monitoring of the facility, development of the PDT dashboard, and validation of the research findings for real-life applications to make a sustainable smart

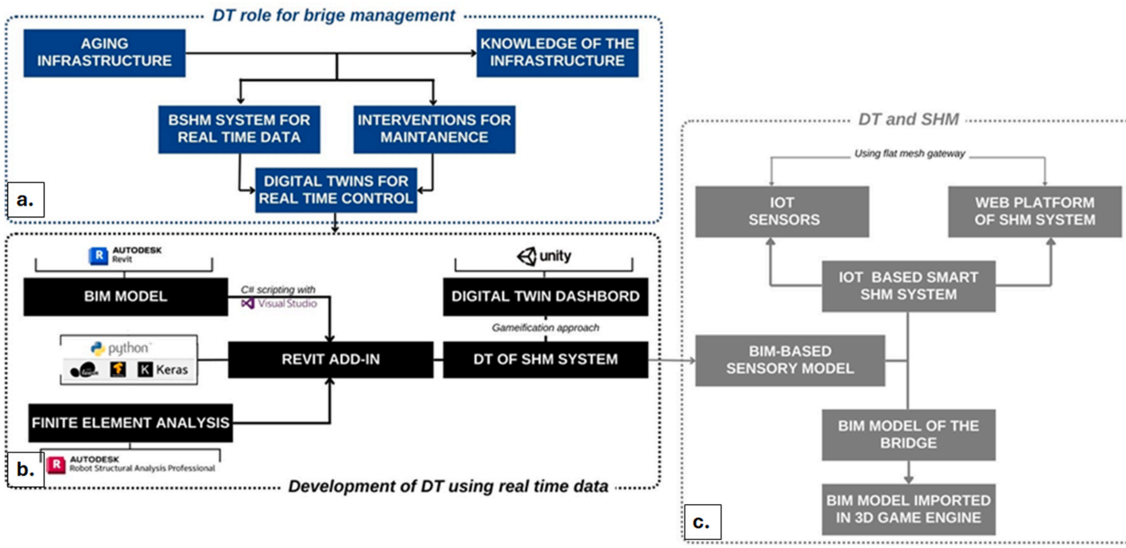


Fig. 1. The role of PDT in the framework of infrastructure management.

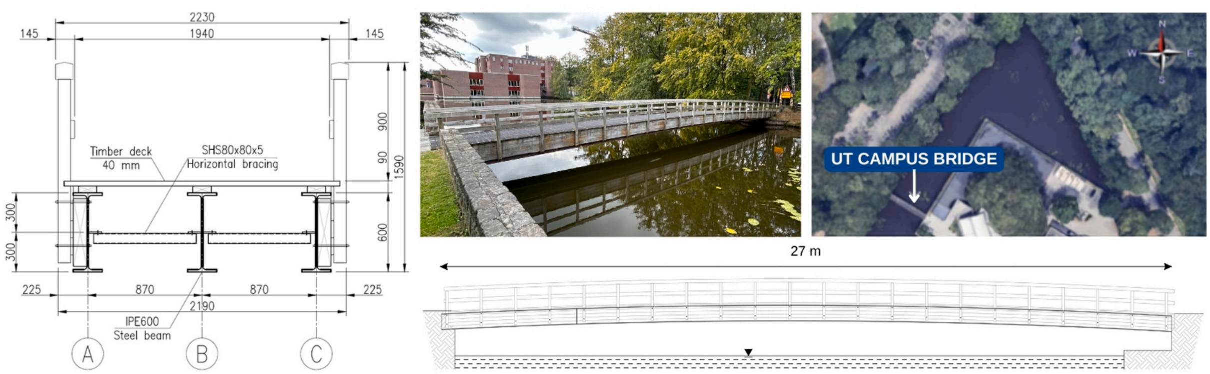


Fig. 2. Cross-section and longitudinal section of the bridge, together with a photograph of the structure and an overview image showing its location and orientation within the surrounding environment.

infrastructure.

Several strain gauges installed along the main girders are used to capture strain responses under operational loading conditions. These sensors are positioned at selected locations along the bridge span and provide the strain measurements used as inputs for the machine learning model in the proposed PDT framework. The monitoring system continuously records the structural response of the bridge, allowing the collected measurements to be used for both model calibration and validation.

6. Development of a predictive digital twin (PDT) platform for the bridge

This research integrates Building Information Modeling (BIM), Finite Element Analysis (FEA), and Digital Twin (DT) technologies for visualization and validation of the performance of the structural system. The research incorporates a BIM model of the bridge, which is provided with the Revit add-in developed by a C# script. The add-in helps to integrate the results of bridge SHM data and compare them with the Finite Element (FE) results in the PDT framework to perform automated bridge health monitoring.

The PDT framework is implemented as a Revit add-in developed from scratch using the C# object-oriented programming language, which forms the basis of the Revit API. The tool runs as a Revit add-in, accessible from the software ribbon, and is responsible for managing the continuous bidirectional data exchange between the physical structure and its digital counterpart, thereby enabling the monitoring and inference of the correct structural behavior of the bridge. The general architecture of the proposed solution is shown in Fig. 3.

The physical structure is monitored by a system of sensors that records its response to the applied loads, primarily the strain and acceleration at the designated measurement points. In the context of the proposed framework, the measurements obtained from the strain gauges are particularly crucial. Among these, two groups are distinguished: one used directly for load estimation and a control group, recording strain responses for the purpose of the comparative analysis conducted at the end of the described procedure. The structural response, measured by the SHM system in the operational stage, is utilized within the ML-based framework to estimate both the magnitude and spatial distribution of live loads. Once the active load has been identified, it can be reintroduced into the FE model to predict the structural response in other parts of the structure, including control measurement points. The actual strain values recorded physically within the control group are then compared with the expected values derived from the FEA. Any deviations between the values may indicate anomalies and, consequently, potentially abnormal structural behavior. The process of data exchange between the physical object and its corresponding digital representation originates from the BIM model. The developed add-in is configured to acquire and manage real-time sensor data, perform load estimation, run the comparative FEA, and visualize the results directly within the BIM

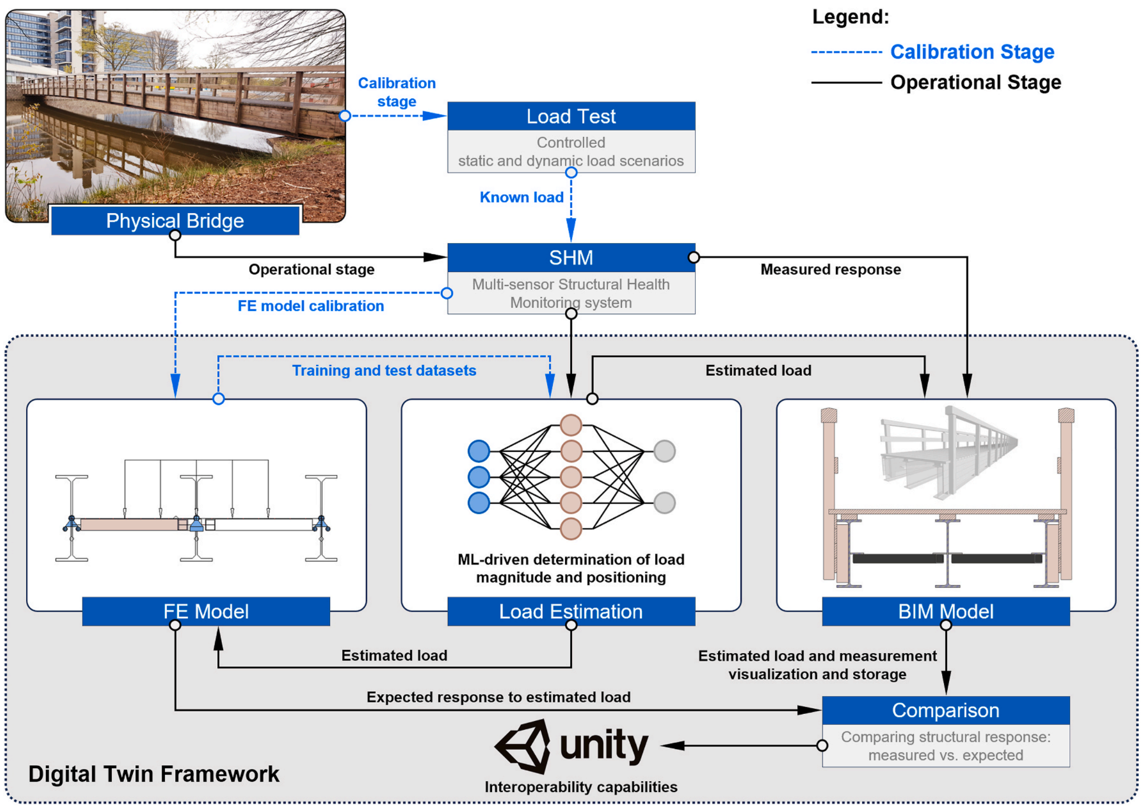


Fig. 3. Overview of the PDT framework.

environment, specifically the estimated loads and both actual and simulated responses. Thanks to the use of a BIM-oriented solution with high interoperability capabilities, this information can be stored and used outside the BIM model, for example, incorporating VR and AR techniques.

However, it should be noted that the described approach would not be possible and reliable without a calibrated FE model. The critical stage in the calibration process involves the analysis of the signals acquired by the SHM system during loading under strictly controlled conditions and, subsequently, attempting to reproduce the observed response in the FE model. The procedure for adjusting and preparing the framework for the operational stage is given in Fig. 3 by the relationships highlighted in blue. In the first step, the bridge is subjected to a static and, optionally, a dynamic load test. In this case, the tests consisted of applying loads of known weight at specific locations. Based on the measured response, a direct comparison can be made with the results obtained from the simulation analysis. FE model calibration involves modifying the stiffness characteristics of the structural elements to minimize the discrepancies between the simulated and real strains. The calibration of the FE model, performed manually or automatically, is the subject of numerous articles (Bedon et al., 2018) and is beyond the scope of current research. The purpose of model calibration is to ensure that the FE model reflects the actual behavior of the structure and to eliminate discrepancies arising from uncertainties, idealizations, and other modeling simplifications. They include the influence of secondary and non-structural elements on the stiffness and load distribution, including the handrails and the wooden deck, clearances, friction, initial deformations, and unknown material properties. Finally, with the calibrated FE model, training and test sets are created to train and validate the neural network that forms the core of the ML-driven load estimation module. The calibration is conducted once in the initial stage and can be reproduced in the case of repetitive anomaly signals under operational conditions.

The development of different elements of the PDT has been discussed in the subsequent sections of this article.

6.1. Virtual model integration workflows

Revit add-ins are DLL libraries that are referenced by manifest files that indicate their exact location on the computer. Once compiled and loaded, an add-in can be called from the Revit ribbon to perform specific operations on a BIM model that will be used as a background for the proposed framework.

The development of the Revit add-in is carried out in Microsoft Visual Studio (VS), where a new project is developed with the C# platform. To gain access to Revit API features, Revit libraries have to be referenced. If the Revit environment is installed, the libraries are available in the Revit root directory by default. So, the C# script uses the RevitAPI.dll and RevitAPIUI.dll libraries from the Revit directory.

An add-in is called from the ribbon in the Revit environment. For successful implementation, it is required to provide a specific code with an execute method that is triggered when a user clicks to open and run the add-in. After this, Autodesk. Revit.Attributes, Autodesk. Revit.DB and Autodesk. Revit.UI libraries are added to the script. It is followed by indicating a transaction type by adding the Transaction attribute above the line with the name of the class. It is not possible to perform operations on the model without the transaction being opened. Only one transaction can be opened at a time. After this, an IExternalCommand interface is added at the end of the line with the name of the class. Then the code is pasted with the default Execute method required by the interface. The Execute method is a three-parameter function that returns a Result enumeration value. The command data, the object of the External CommandData type, stores the information about the Revit application instance and the current document that the add-in is called from.

6.2. Graphical user interface design for BIM integration

A graphical user interface (GUI) is developed using Windows Forms within Visual Studio. The interface includes components such as buttons, textboxes, and file dialog controls that allow users to input data, execute the machine learning model, and visualize results. The interface is designed to be intuitive, providing clear instructions and outputs for the user.

For this purpose, a Windows Form is designed and integrated into the project. This form serves as a graphical user interface (GUI) for the add-in, allowing users to input data, trigger the add-in's functionalities, and visualize results. Controls like buttons, labels, and textboxes are added to the form to capture user input and display outputs. An OpenFileDialog control is included to allow users to browse and select a file containing sensor data. The selected file path is displayed in a form textbox. This functionality is crucial for loading SHM data into the system, which will later be used for ML predictions and FEA. The form is linked to the Revit environment by modifying the Execute method so that the interface is instantiated when the add-in runs. This integration ensures that the user can interact with the BIM model directly through the Revit interface.

6.3. Testing the graphical user interface

Once the add-in code is complete, the project is built using the Build Solution option in Visual Studio. The output is a Dynamic Link Library (DLL) file, which contains the compiled code necessary for the add-in to function within the Revit environment.

A manifest file with the .addin extension is created to provide Revit with the necessary information about the add-in, such as its name, the path to the DLL, and the class that contains the Execute method. This file is placed in the Revit Add-ins directory, allowing Revit to recognize and load the add-ins upon startup.

With the add-in installed, Revit can now load it and present it in the External Tools section of the Add-Ins tab. The user can run the add-in by selecting it from the Revit ribbon, which launches the custom form designed earlier. The form enables users to perform various SHM-related tasks directly within the BIM model. An interface of the BIM model with the add-in is shown in Fig. 4.

6.4. Finite element analysis of the bridge

Finite Element Analysis (FEA) of the bridge is carried out to calculate internal forces and displacement. The data collected from the FEA can then be used to simulate the structural behavior of the bridge and compare it with the real response measured by the SHM system (Faisal et al., 2018).

To carry out the static linear analysis, a linear elastic model of the bridge is developed as a shell and linear element. The model consists of 1D bar elements – the girders and the bracing. The stiffness of the wood deck and non-structural elements on the bridge was neglected. The model was supplemented with a thin and weightless shell that allows the load to be transferred between the girders. The developed FE model of the bridge is shown in Fig. 5.

The FE model is calibrated to reproduce the response of the physical structure under a load of known magnitude and placement. This involves modifying the stiffness characteristics of the model elements, primarily geometric characteristics and elastic moduli, thus reflecting the influence of non-structural elements and various uncertainties. Given the scope of the research, focusing on application aspects, machine learning, and integration of individual PDT modules, the calibration procedure is not presented here in detail.

The accuracy of the process can be validated by analyzing the signals recorded by the SHM system for known loads that are not involved in the calibration process. Fig. 6 presents the processed time series of strain values obtained from three representative sensors installed at different cross-sections along one of the girders. The processing in question allows signal compression and reduces the impact of noise and dynamic effects. The authors propose the use of an Infinite Impulse Response (IIR) low-pass Butterworth filter, modifying the signal forward and backward to achieve zero-phase distortion, although the design of the filter is not critical in the current study and may be replaced with other methods and configurations suitable for specific purposes. The signal is then compressed, which records the minimum and maximum values in 5s windows, thus reducing the resources required for data storage and visualization within the PDT. In Fig. 6, the influence of temperature on the measured strain values at the measurement points may be observed. However, given the transient character of the operational loads and focusing on strain increase in short periods, additional processing, including temperature compensation, is not required.

The calibrated FE model is used to simulate the structural response of the bridge to the applied loads, including a variety of scenarios that cannot be performed on a real structure due to time and resource constraints. This response is characterized by the strains and stresses at specific nodes, which correspond to the locations of the SHM sensors on the actual bridge. The simulated data provide a baseline for comparison with real-world measurements.

A uniform load of dimensions 1.0×1.0 m and a magnitude of 1.0 kN/m^2 was added to the model at different locations along the bridge, every 0.25 m. After carrying out the FE analysis, an Excel file containing a worksheet with raw FE output indicating the element, node, and placement of the load along the bridge is generated. The results of axial force and bending moments in a given node, as per the FE analysis, are added to the same file. Based on their values, the geometric characteristics of the girders' cross-section, and Young's modulus of the steel, stress and strain can be calculated directly. Similarly, another worksheet of the ANN dataset contains randomized data for different placements and load multipliers in each row. Finally, records were generated and copied to the text file. Its content will be loaded and used for the training of an ML regression model. The first five columns in the text file indicate scaled strains, while the last two columns indicate load placement and weight, respectively.

The results of the FE analysis are integrated into the Revit add-in,

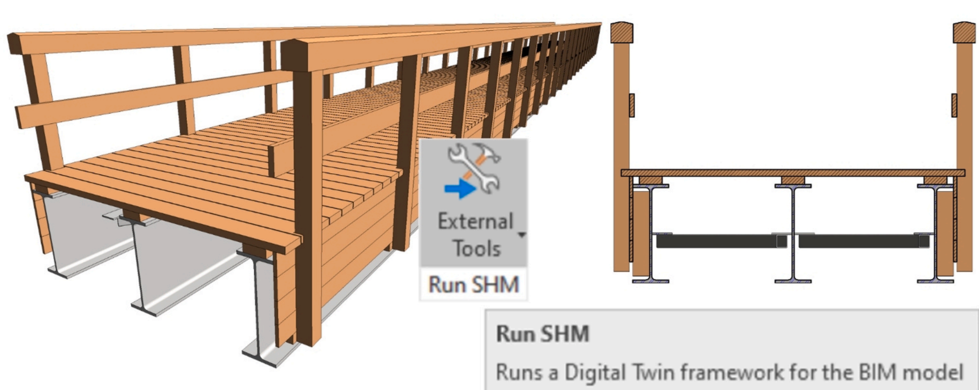


Fig. 4. The interface of the BIM model with the Revit add-in.

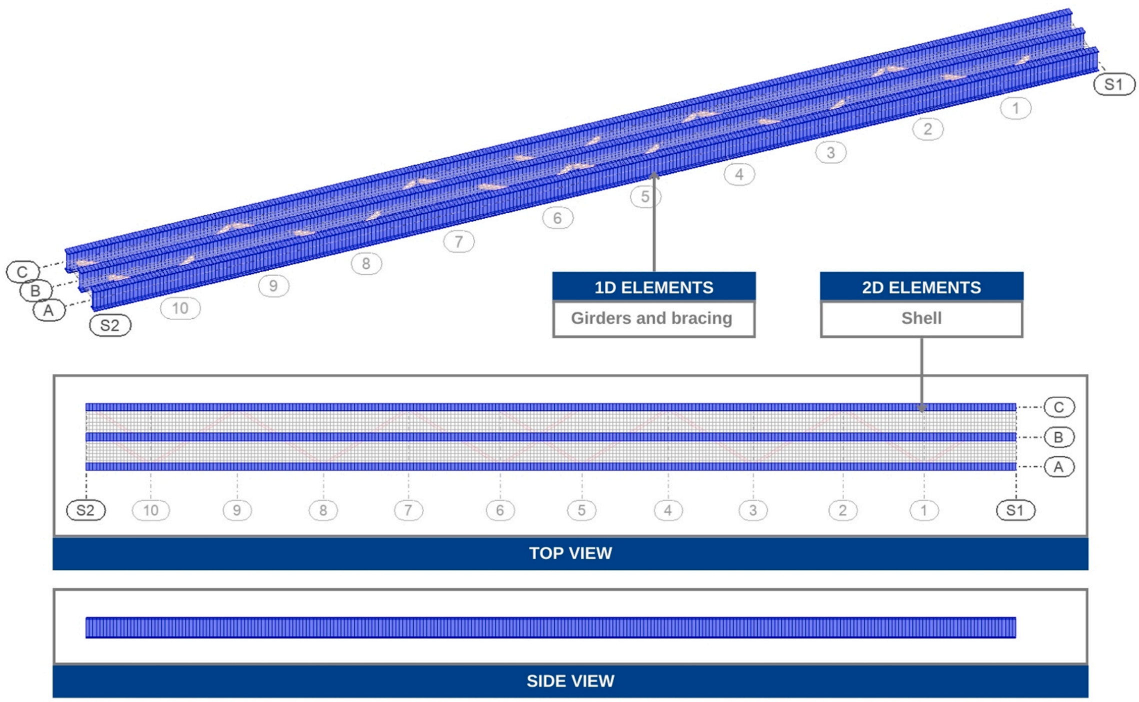


Fig. 5. Finite Element (FE) model of the bridge used for the structural analysis.

enabling real-time visualization of the bridge's structural behavior within the BIM model. This integration allows for continuous monitoring and comparison of simulated and actual data, providing critical insights into the health of the structure.

6.5. Integration of the FEA with the ML dataset

In the proposed framework, the FE model is used to simulate the structural response of the bridge under different load scenarios and generate datasets linking strain responses with load characteristics. The ML model is then trained on these FE-generated datasets to learn the relationship between sensor measurements and load placement and magnitude. A dataset is prepared for training a machine learning model. The dataset includes input-output pairs derived from the FE simulations, where the inputs are strain measurements from selected nodes, and the outputs are the corresponding load placement and magnitude. This dataset is formatted and stored in a CSV file, ready for use in ML training. The dataset along with different steps of ML algorithm development is shown in Fig. 7.

The first 15 lines of the data used for training the ML algorithm are shown in Fig. 7. The last 7 columns contain strain measurements at the locations of the five selected sensors, as well as the load placement and its weight. Thus, columns 4-8 are the inputs of the ML algorithm, and columns 9 and 10 are its expected outputs. The CSV file was first parsed to extract the required data from it. The dataset is then randomly divided into training and test subsets. These datasets were then used to train a simple ML model.

The selected model was a fully connected ANN implemented using Keras. The input features to the network were the strain measurements obtained from five sensors, resulting in an input shape of (n, 5). The model outputs were load placement and load weight, resulting in an output shape of (n, 2). The adopted network was a fully connected neural network composed of three dense layers with 64, 32, and 2 nodes, respectively, where the final layer generated the two target outputs. Dense layers in Keras correspond to standard fully connected layers. The first and second hidden layers used the ReLU (Rectified Linear Unit) activation function, while the output layer used a linear activation function. For model development, the available dataset was divided into 80% training data and 20% validation data. The model was trained using the mean squared error (MSE) loss function and the Adam optimizer with default parameter values, which are learning rate = 0.001, β1 = 0.9, β2 = 0.999, and ε = 1e-7. A batch size of 32 was used during training. The training process was continued for 790 epochs, after which the loss had converged, and further training did not lead to noticeable improvement in model performance.

This model was chosen because the main goal of the study was to integrate a reliable ML module into the DT framework rather than to explore increasingly complex ML architectures. In the present application, the input space is limited to strain measurements from five sensors, and the outputs are two target variables, namely load placement and load weight. Given this relatively low-dimensional regression problem, together with the modest structural complexity of the bridge under study and its FE model, a compact fully connected ANN was considered sufficient to learn the required nonlinear mapping, as supported by the obtained test results. More complex alternatives, such as deeper or more specialized neural network architectures, were not adopted because they would increase computational and implementation complexity without a clear expected benefit for this problem setting. Therefore, their use was not considered justified by the complexity of the problem.

To assess deployment feasibility, we evaluated the computational performance of the proposed system on a Lenovo ThinkPad P15v Gen 3 mobile workstation equipped with an NVIDIA T1200 Laptop GPU (4 GB GDDR6) and 32 GB DDR5 RAM. The average inference time of the proposed model was approximately 60 ms per step, which corresponds to about 16.7 inferences per second. This result suggests that the method is suitable for near-real-time operation on a practical mobile-workstation platform. It should be noted that the reported value reflects model inference time; the total end-to-end latency in a deployed system may be higher.

The results of this model are presented in Table 1. The MSE, mean absolute error (MAE), and coefficient of determination (R^2) were used to evaluate the model. MAE shows the average magnitude of error in the same unit as the variable. Moreover, R^2 shows the capability of the model to explain the variance in the variable, where one is a perfect

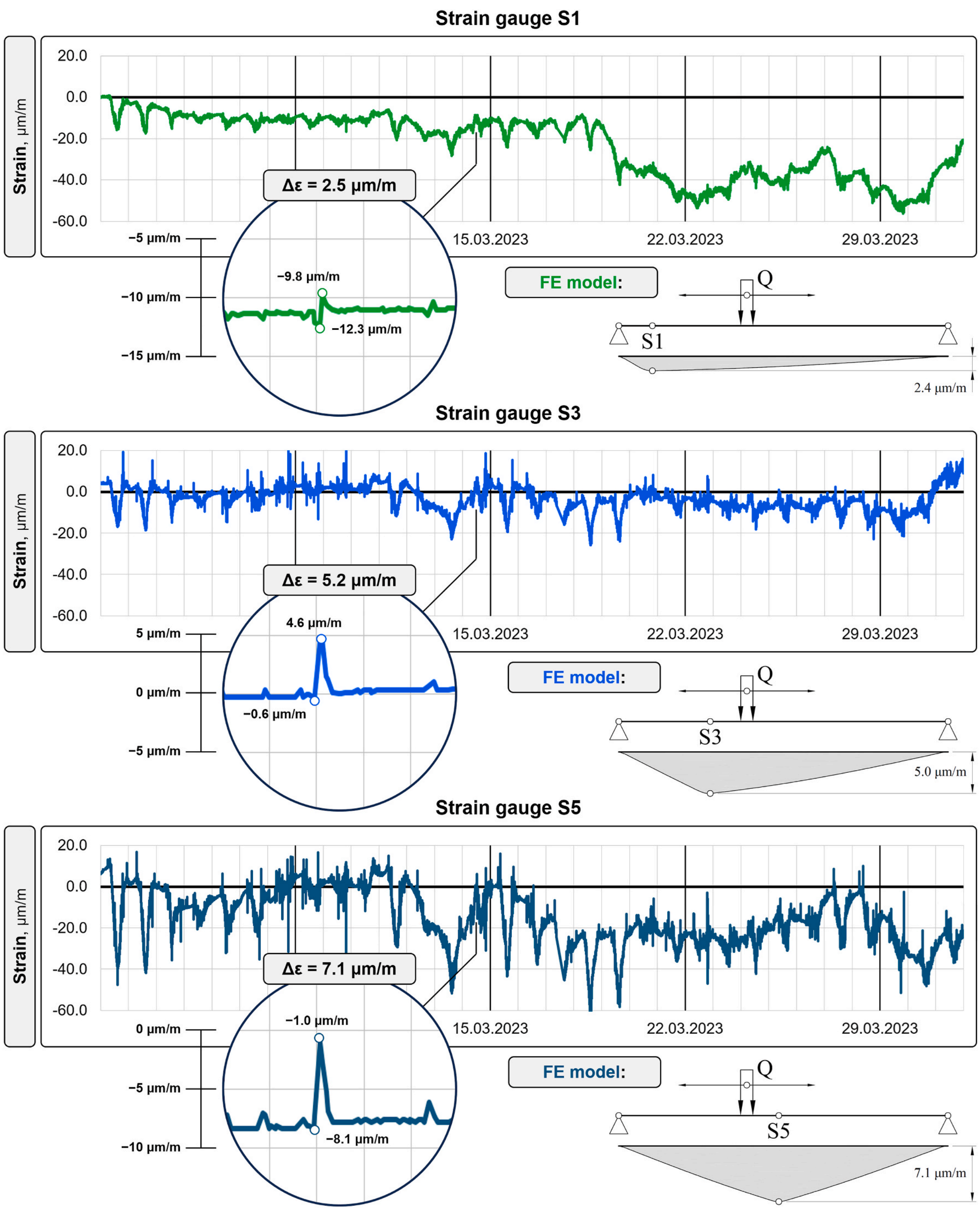


Fig. 6. Strain time-series and comparison of live load effects between real-world measurements and FE-based simulations.

representation. Fig. 8 presents the predicted values compared to the actual values in a test set.

The aim of this study is to demonstrate the feasibility of integrating FEM, ML, and BIM within a PDT framework rather than benchmarking multiple modeling approaches. In this context, the FE model serves as a physics-based baseline for generating the dataset and validating the ML predictions. Future work will include systematic comparisons with alternative approaches such as FEM-only simulations or traditional BSHM methods.

Fig. 8(a) shows that the predicted values for load placement exhibit a slight tendency to lie above the 45-degree reference line, indicating a minor positive bias in the model predictions. This suggests that, in some cases, the model slightly overestimates the load position. Such behavior may result from the spatial resolution of the FE-generated dataset and the sensitivity of strain responses to nearby load locations, where small variations in strain can correspond to adjacent placement positions. Despite this minor bias, the deviation is limited, as reflected by the high coefficient of determination ($R^2 = 0.9757$) and low error values, confirming the strong predictive performance of the proposed framework.

The trained neural network model is integrated into the Revit add-in through a C# script that calls a Python executable. This script loads the neural network model, processes the input data, and returns predictions that can be visualized within the BIM model. The predicted load characteristics are then used to simulate the structural response in the FE

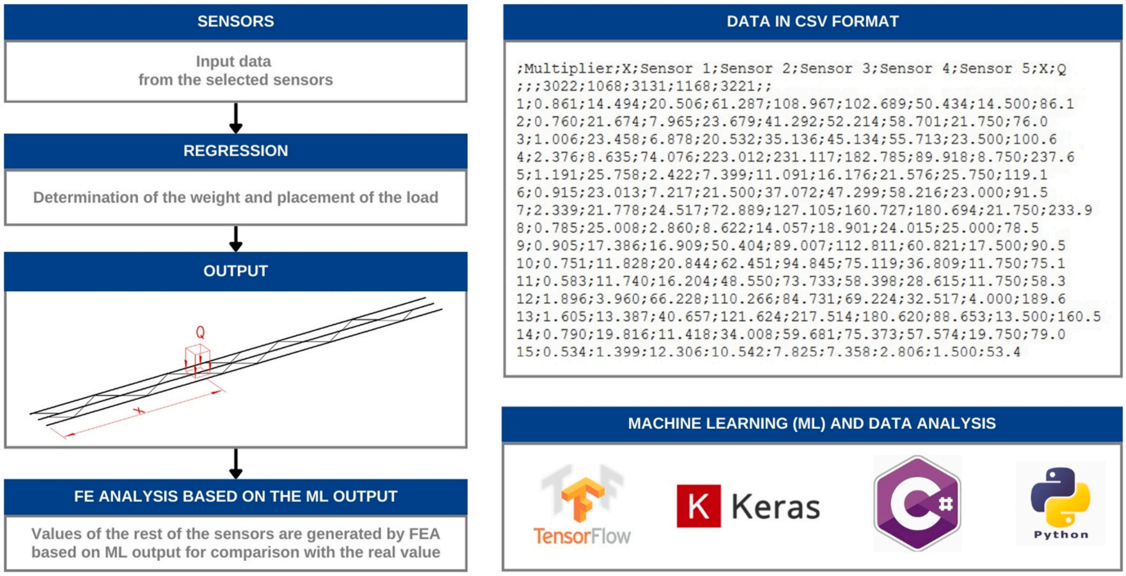


Fig. 7. The flowchart of the ML process and part of the dataset generated by the FE model for training the ML model.

Table 1
The results on the accuracy of the PTD model at the test phase.

Metric	Load Placement	Magnitude
MSE	1.2324	0.8983
MAE	0.9445	0.6399
R^2	0.9757	0.9997

model, completing the loop between real-world measurements, ML predictions, and structural analysis.

6.6. Validation of the predictive digital twin on the Bridge

The add-in is tested using real-world SHM data collected from the bridge. The predicted load characteristics are compared against test load cases of known magnitude and placement to validate the accuracy of the neural network model. The test scenarios include field experiments conducted under controlled conditions and allow for the collection of strain data resulting from specific loads that are not incorporated in the calibration and regression model training procedures.

It is made sure that the file textbox provides a correct path to the sensorsdata.txt file. When the Run button is clicked, the program performs the analysis. The predicted load parameters are printed to the text boxes in the middle. The simulated measurements of the rest of the sensors are printed in the boxes on the right side of the form.

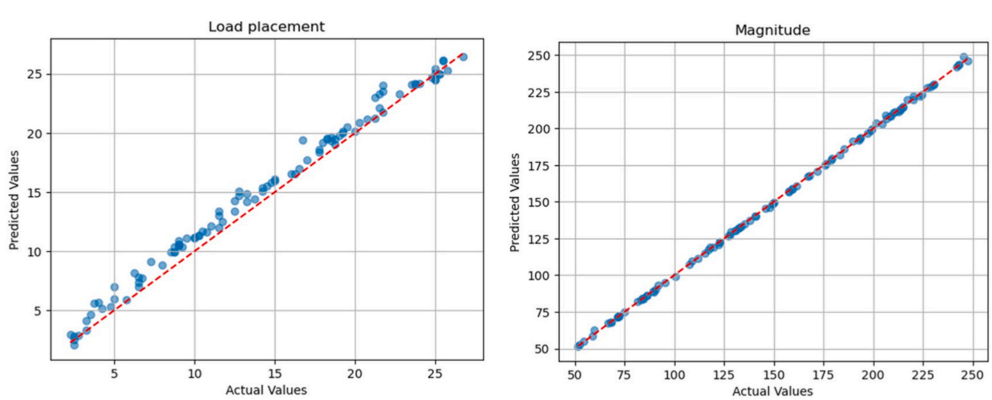
The simulated structural response from the FE model is compared with actual measurements obtained from the SHM system. Discrepancies between the simulated and measured data are analyzed to identify potential anomalies in the bridge's behavior.

The PDT framework, as shown in Fig. 9, undergoes iterative refinement based on the validation results. The ML model and FE simulations are adjusted as necessary to improve accuracy and reliability. Once validated, the DT can be used for continuous monitoring, providing real-time insights into the structural health of the bridge.

The proposed approach provides a comprehensive approach to developing a digital twin dashboard for structural health monitoring. The integration of machine learning models, finite element analysis, and BIM visualization tools offers a powerful system for predicting, monitoring, and assessing the structural integrity of bridges.

6.7. Development of the bridge PDT dashboard using a gamification approach

A novel gamification technique is used to develop the PDT model of the bridge SHM system. For this purpose, a 3D game (UNITY 3D) engine is used. The benefits of using the 3D game engine are not only to develop the PDT dashboard of the bridge SHM system, but the visualization and



(a) The placement of the load on the bridge

(b) The magnitude of the load on the bridge

Fig. 8. The placement and magnitude of the load predicted by the ANN model versus the actual values of a test set are shown in (a) and (b), respectively.

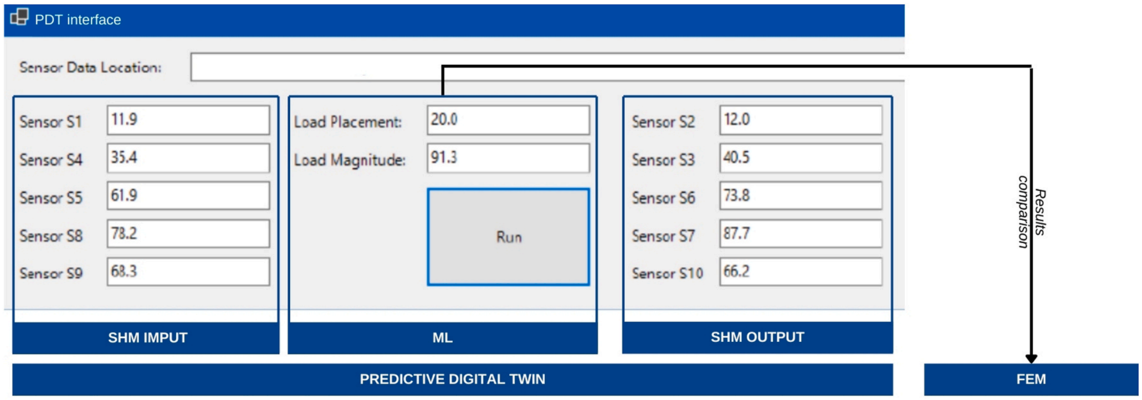


Fig. 9. Digital Interface developed in Revit.

management of SHM data is also seamless through the developed dashboard, which provides the possibilities of proactive decision-making and on-site bridge health monitoring. The schematic diagram of the steps followed in the creation of the PDT framework using the gamification approach is shown in Fig. 1(c).

The next step is the creation of a BIMified sensory model, which is developed by deploying the smart sensors to the BIM model of the bridge. This process is implemented using the UNITY platform, which only imports the FBX file formats of the BIM model. The BIM model is therefore exported in FBX format and imported into the 3D game engine. It's worth mentioning that the IFC formats or.rvt files can also be used with third-party 3D animation apps. Once the BIM model is imported into the game engine, twins of physical sensors are generated considering their actual locations. In this way, virtual sensor twins replicate the functionality of the physical sensors within the PDT environment.

Once the virtual sensor twins are created, the next step is to establish real-time communication between the virtual and physical sensors. Each virtual sensor is therefore equipped with automation functions within the gaming environment. Canvases are used to mesh the PDT model that creates regenerative algorithms and connects them to the PDT model in order to establish communication between the twin and real sensors. These meshes establish a direct connection with the IoT-web platform of the SHM system and regulate the augmentation of the virtual elements that are generated. A unique C# script is created in Visual Studio and then embedded into the canvas to automate the communication between the PDT model and the canvas. This initiates an automatic connection between the SHM system's IoT web platform and the PDT model. Twin sensors are represented by clickable buttons within the PDT model. When activated, the embedded C# algorithms establish communication between the virtual and physical sensors. In this approach, the entire twin system is created from scratch, which can interact with the physical SHM system automatically and carry out its operations with a single click of virtual buttons, creating the digital twin of the installed SHM system. The dashboard of the developed PDT model and its integration with the physically installed SHM system are shown in Fig. 10.

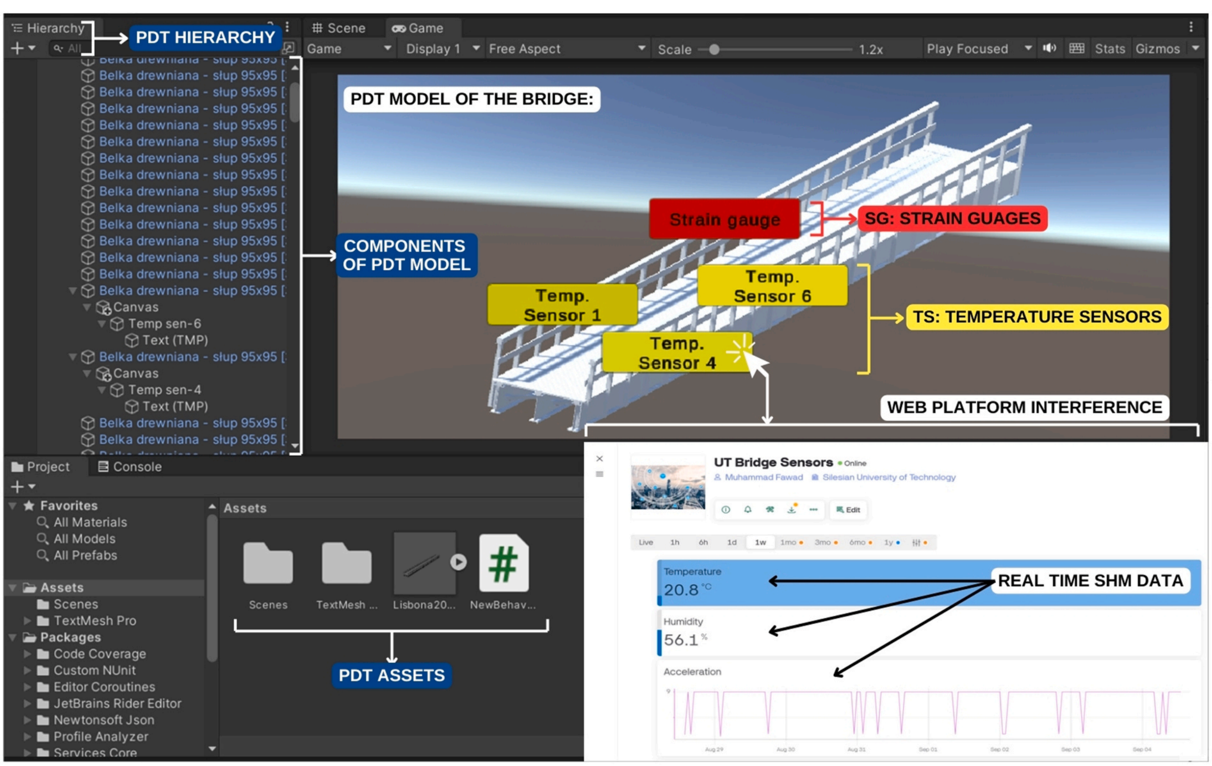


Fig. 10. Bridge PDT dashboard in a 3D game engine.

7. Discussion

This research presents the successful development and implementation of a comprehensive Predictive Digital Twin (PDT) dashboard for the Structural Health Monitoring (SHM) control of the bridge presented as a case study. The integration of Building Information Modeling (BIM), Finite Element Analysis (FEA), and Machine Learning (ML) has been pivotal in providing real-time insights into the structural integrity and behavior of the bridge. By synthesizing these advanced technologies, the PDT framework offers a dynamic platform that facilitates continuous monitoring, predictive analysis, and informed decision-making regarding the structural integrity of the bridge asset. The holistic approach has demonstrated its capacity to simulate the bridge's structural responses accurately under various loading conditions, which is indispensable for proactive maintenance strategies and safety assessments.

The findings of this study align with and expand upon the existing body of research, highlighting the considerable benefits of implementing Digital Twins in infrastructure monitoring. The combination of BIM and FEA within the PDT framework not only enhances the precision of structural behavior predictions but also reinforces the ability to monitor the condition of the structure in a way that is both data-driven and automated. Specifically, the FEA component allows for a detailed simulation of stress and strain responses across different load scenarios, ensuring that the virtual model can replicate the behavior of the physical structure with a high degree of accuracy. This capability is essential for identifying potential issues before they manifest in the real world, thereby enabling timely maintenance interventions.

Moreover, the incorporation of machine learning models into the PDT framework represents a significant advancement in the field of SHM. By leveraging sensor data, the ML models employed in this study can predict load characteristics, such as stress distribution and deflection patterns, with a high level of precision. This not only enhances the accuracy of the structural simulations but also streamlines the process of decision-making by automating the interpretation of complex datasets. Thus, the integration of ML algorithms provides deeper insight into structural behavior under operational conditions and enables continuous improvement of the SHM system through refined predictions. This data-driven approach marks a notable progression towards more intelligent and autonomous SHM solutions, thereby contributing to the state of the art in this domain.

However, this research is not without its limitations. One significant limitation is that the investigated bridge is relatively small, which may limit the generalizability of the findings to larger or more complex infrastructures. Bridges of different scales or designs may exhibit distinct structural behaviors, and the current PDT framework may need adaptation or further validation before it can be applied universally across various types of infrastructures. Additionally, the performance of the machine learning model is highly contingent on the quality and volume of sensor data available. Variability in the environmental conditions, sensor placement, and data acquisition methods could impact the accuracy of the ML predictions, especially in more diverse or challenging operational settings.

In light of these limitations, further research should aim to extend the applicability of this PDT framework by testing it across a broader range of bridge types, including those with more complex geometries or load-bearing characteristics. Furthermore, there is considerable scope for improving the predictive accuracy of the machine learning models by integrating more sophisticated ML techniques, such as deep learning or ensemble methods, capturing the complexity of the data, and providing more robust predictions. Investigating the potential of hybrid models that combine traditional FEA with advanced ML techniques could also be a promising avenue for enhancing the predictive power and reliability of the PDT framework in future studies.

8. Conclusion

This research developed a detailed Predictive Digital Twin (PDT) Dashboard for the Structural Health Monitoring (SHM) of a steel girder bridge located on the University of Twente Campus, Netherlands. By integrating Building Information Modeling (BIM), Finite Element Analysis (FEA), and Machine Learning (ML) into a unified framework, the study demonstrates how advanced systems work together to provide real-time insights into structural performance, along with monitoring the integrity of the case study.

Key findings of this research include the successful integration of the BIM model of the bridge with the real-world SHM data carried out through a PDT-based Revit add-on developed using the C# algorithm. The PDT model enables real-time visualization of SHM sensor data together with FEA simulation results within the BIM environment. The FEA enables the calculation of strain, stress, and deflection patterns under various load conditions. The analysis provides detailed simulations of the structural responses. The PDT model shows that the FE results closely align with real-world measurements obtained from the SHM system.

In the case of the structural aspects, analysis was limited to a static approach. Although pedestrian and cyclist traffic is dynamic and random in nature, no extreme events (e.g., synchronous structural excitation) that would necessitate a departure from static assumptions were recorded during the monitoring period. Considering typical service conditions and the low mass of live loads, a quasi-static approach was adopted. This is sufficient for the logical verification of the interface and model calibration, although a full dynamic analysis could provide a valuable extension for future research. Moreover, the FEM computational model contains intentional simplifications regarding geometry and loading parameters. However, it should be emphasized that the FEM model meets the definition of a digital twin, as it has been subjected to rigorous calibration based on actual SHM data. Consequently, any stiffness resulting from the presence of elements omitted in geometric modeling was indirectly accounted for in the model's global stiffness parameters. The use of a massless shell allowed for a realistic load distribution without artificially stiffening the structure. The adopted assumptions are consistent with standard engineering practice for bridge load testing, where the structural response is dominated by the main steel system.

Furthermore, the article uses a Machine Learning (ML) model implemented using Python-based libraries. The study integrates it with the PDT model, which provides accurate predictions of the load placement and magnitude based on strain measurements obtained from the installed sensors. Although a simple neural network model was used in this research, the predictions were consistent with the known load cases. This validates the use of ML in the context of bridge health monitoring, demonstrating its capacity and capability to process real-time data and provide predictive insights. These insights can be used to support proactive maintenance strategies.

For better visualization of the developed PDT framework and to perform real-time SHM of the infrastructure, this research also introduced a novel gamification approach by utilizing a 3D game engine to create an immersive form of a PDT dashboard. The dashboard offers seamless interaction and visualization of the SHM data. The real-time integration between physical sensors and their digital counterparts within the game environment provides a unique yet useful method for monitoring and managing infrastructural health.

Conclusively, this research provides a useful framework by integrating BIM, FEA, and ML into a PDT framework of the bridge SHM system, offering a real-time data-driven solution for monitoring and maintenance of the infrastructure. The successful application of this framework on the UT campus footbridge demonstrates its potential for larger use in smart infrastructural management. Future expansions and improvements of this work can further contribute to the development of autonomous systems for infrastructural health monitoring.

CRediT authorship contribution statement

Marcin Jasiński: Conceptualization, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Writing – original draft. **Muhammad Fawad:** Conceptualization, Formal analysis, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ali Sabzi Khoshraftar:** Formal analysis, Methodology, Resources, Software, Validation, Writing – original draft. **Alessia Abbozzo:** Investigation, Methodology, Software, Visualization, Writing – original draft. **Yongjian Tao:** Data curation, Methodology, Software, Validation, Writing – original draft. **Marek Salamak:** Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Borys Kopeć:** Investigation, Methodology, Software, Writing – original draft. **Qian Chen:** Conceptualization, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors whose names are listed for this paper declare no competing and/or no conflicting interests.

Data availability

Data will be made available on request.

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